

EXPERIMENT – STATIC WEBSITE USING DOCKER

CREATE, CONTAINERIZE, BUILD AND SERVE A STATIC WEBSITE

Aim: To create a simple static website, containerize it using Docker, build a Docker image, and serve the website using an Nginx web server.

Requirements

- HTML file (index.html)
- Docker installed on the system
- Dockerfile
- Nginx web server image
- Web browser for testing

Procedure

- 1. Create the Static Website:** Create an **index.html** file containing basic HTML content and save it in the project directory.
- 2. Create the Dockerfile:** Configure Nginx to serve the website by copying index.html into the Nginx HTML directory.
- 3. Build the Docker Image:** Build the image from the Dockerfile using the Docker build command.
- 4. Run the Container:** Start a Docker container and map a host port to the Nginx container port.
- 5. Access the Website:** Open a browser and visit the mapped localhost port to verify that the static website is served successfully.

Dockerfile

A typical Dockerfile for serving the static website with Nginx is:

```
FROM nginx:latest
COPY index.html /usr/share/nginx/html/index.html
EXPOSE 80
```

Docker Commands

Build the image:

```
docker build -t static-website .
```

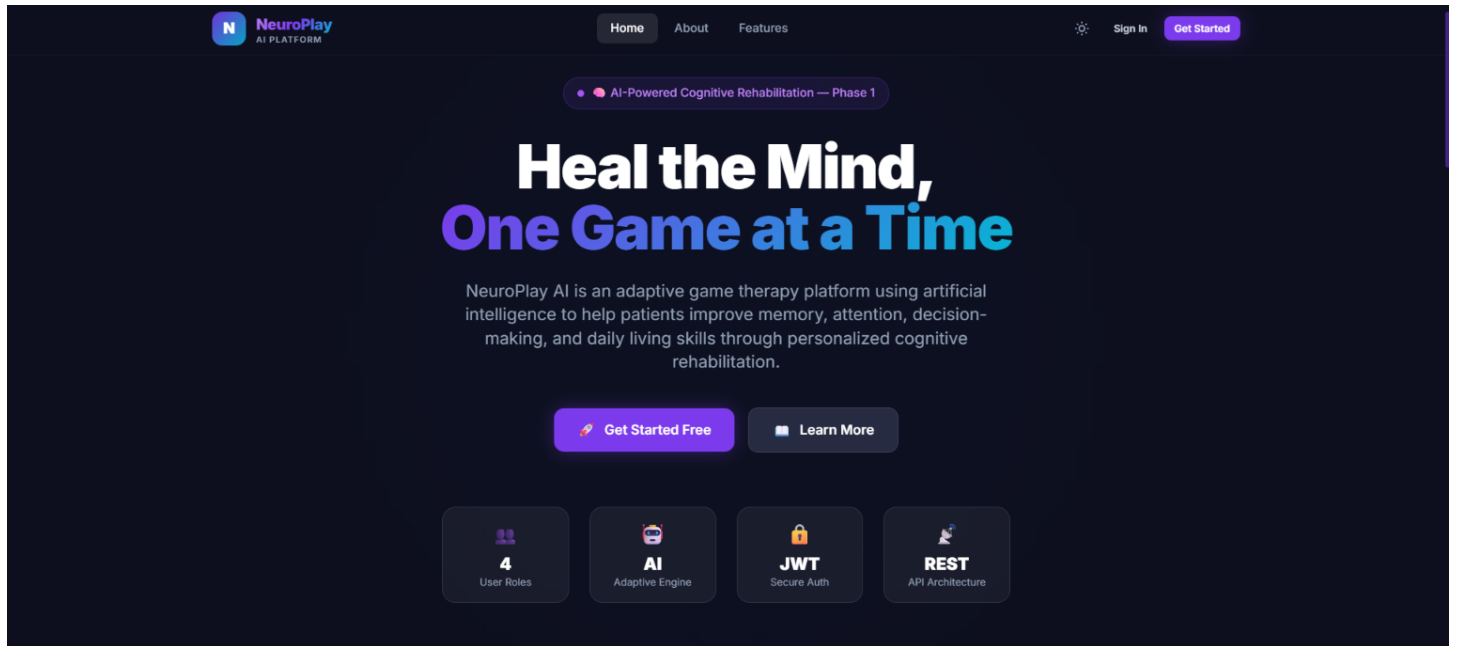
Run the container:

```
docker run -d -p 8080:80 static-website
```

Result

The static website was successfully containerized using Docker and served through Nginx. The website was accessed through a web browser, confirming that the Docker image and container were working correctly.

SCREENSHOT OF COMPLETED STATIC WEBSITE



Conclusion: The website was created as a static HTML page, containerized using Docker, built into an image, and successfully served using Nginx inside a Docker container.